Lecture 8

A particle on a ring of N sites

Objectives. *A particle that can sit on any of N sites arranged in a ring, and hop between neighbouring sites, is the first system in this course with a position. Lecture 6 showed from its symmetries alone that it has a degenerate level. This lecture finds all its levels. The shift that moves the particle round the ring is diagonal in the basis of waves around the ring, the discrete Fourier basis, and so is the Hamiltonian. Benzene, in a simple model, is the ring of six sites. A particle started on one site spreads round the ring and, for six sites, comes back. With two sites in each cell and unequal alternating hops, the levels form two bands separated by a gap.*

By the end of this lecture you will be able to:

- Find the levels of a ring and their degeneracies from the Fourier basis.
- Find the probability of each site at the time t for a particle started on one site.
- Find the levels of a ring with a flux, and the flux at which two of them meet.
- Find the two bands of a ring with two sites in each cell, and the gap between them.

8.1 A particle that hops

Take $N \geq 3$ sites on a ring, labelled $x \in \mathbb{Z}_N$, so that the site $x + N$ is the site x. A particle sits on one of them, and the state in which it sits on the site x is the unit vector e_x. The vectors e_x are an orthonormal basis of $\mathbb{C}^N$, and a measurement in this basis tells which site the particle is on. If the particle could not move, every e_x would have the same energy c, the *on-site energy*. A particle that can *hop* to a neighbouring site has the Hamiltonian

$$H = c\mathbb{1} - \kappa\big(T + T^\dagger\big), \qquad \kappa > 0, \tag{8.1}$$

where T is the shift of § 6.7, $Te_x = e_{x+1}$, and $T^\dagger = T^{-1}$ shifts back. In the basis e_x, the matrix of H has c on the diagonal, $-\kappa$ in the entries that join neighbouring sites, and 0 elsewhere. The number κ, the *hopping amplitude*, sets how fast the particle moves. With $-\kappa$ in front and $\kappa > 0$, the state $N^{-1/2}\sum_x e_x$, with equal amplitudes on all the sites, has the lowest energy, as § 8.3 shows. This is the *tight-binding model* of a particle on a ring.

Lecture 6 found two symmetries of H: the shift T and the mirror M, $Me_x = e_{-x}$. They do not commute, so H has a degenerate level (Theorem 6.5). Which level is it, and are there others?

8.2 The Fourier basis

A symmetry whose eigenvalues are all distinct would settle the question at once. If $Vv = \alpha v$ for a symmetry V, then $V(Hv) = H(Vv) = \alpha Hv$. So H maps each eigenspace of such a

symmetry into itself. Each eigenspace is a line, so every eigenvector of the symmetry is an eigenvector of H. The shift T is such a symmetry. Put $\zeta = e^{2\pi i/N}$, and for $k \in \mathbb{Z}_N$ let

$$f_k = \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} \zeta^{kx} e_x. \tag{8.2}$$

The amplitude of f_k on the site x is $\zeta^{kx}/\sqrt{N}$. Its phase advances by $2\pi k/N$ from each site to the next, so f_k is a wave that winds k times round the ring. Since $\zeta^N = 1$, the vectors f_k and f_{k+N} are the same, and k is a label in $\mathbb{Z}_N$. The vectors f_k are orthonormal. For k and l in $\{0, \dots, N-1\}$, the geometric sum gives

$$\langle f_l, f_k \rangle = \frac{1}{N} \sum_{x=0}^{N-1} \zeta^{(k-l)x} = \delta_{kl},$$

since $\zeta^{k-l} \neq 1$ for $k \neq l$. There are N of them, so they are an orthonormal basis, the *Fourier basis* of $\mathbb{C}^N$.

The shift acts on f_k by a phase. Indeed,

$$T f_k = \frac{1}{\sqrt{N}} \sum_x \zeta^{kx} e_{x+1} = \frac{1}{\sqrt{N}} \sum_x \zeta^{k(x-1)} e_x = \zeta^{-k} f_k,$$

where the second sum renames $x+1$ as x. So f_k is an eigenvector of T with the eigenvalue ζ^{-k}. The minus sign comes from moving the particle forward. The N eigenvalues ζ^{-k} are distinct, so every eigenspace of T is a line.

8.3 The levels of the ring

Theorem 8.1 (The levels of the ring). *Let $N \geq 3$ and let $H = c\mathbb{1} - \kappa(T + T^\dagger)$ with c real and $\kappa > 0$. Then $Hf_k = \varepsilon_k f_k$ for every $k \in \mathbb{Z}_N$, with*

$$\varepsilon_k = c - 2\kappa \cos \frac{2\pi k}{N}. \tag{8.3}$$

Two labels have the same energy, $\varepsilon_k = \varepsilon_l$, if and only if $l = \pm k$ in $\mathbb{Z}_N$. So $k = 0$ is a simple level, and so is $k = N/2$ when N is even. Every other level is twice degenerate, with the eigenspace spanned by f_k and f_{-k}. The mirror exchanges them, $Mf_k = f_{-k}$.

Proof. Since $T^\dagger = T^{-1}$, $T^\dagger f_k = \zeta^k f_k$. So $Hf_k = \big(c - \kappa(\zeta^{-k} + \zeta^k)\big) f_k$, and $\zeta^{-k} + \zeta^k = 2\cos(2\pi k/N)$. Next, choose the representative of each label with $|k| \leq N/2$, so that $2\pi\,|k|\,/N$ lies in $[0, \pi]$. The cosine is strictly decreasing on $[0, \pi]$, so $\varepsilon_k = \varepsilon_l$ exactly when $|k| = |l|$, that is, $l = \pm k$. The labels k and $-k$ are the same exactly when $2k = 0$ in $\mathbb{Z}_N$, that is, $k = 0$ or $k = N/2$. Finally, $Mf_k = N^{-1/2} \sum_x \zeta^{kx} e_{-x} = N^{-1/2} \sum_x \zeta^{-kx} e_x = f_{-k}$. ∎

The levels run from $c - 2\kappa$, for the uniform state f_0, to at most $c + 2\kappa$. The degeneracy that Lecture 6 predicted is the pairing of k with $-k$. The waves that wind the two ways round the ring have the same energy, and the mirror turns one into the other.

Every f_k is spread evenly over the ring, since $|\langle e_x, f_k \rangle|^2 = 1/N$ for every site x. A stationary state need not be spread evenly, however. Inside a degenerate level, every vector is stationary (Theorem 4.4), and a combination of f_k and f_{-k} can vanish on some sites. For example, take $N = 4$. Then

$$f_1 = \frac{1}{2}(e_0 + ie_1 - e_2 - ie_3), \qquad f_3 = f_{-1} = \frac{1}{2}(e_0 - ie_1 - e_2 + ie_3),$$

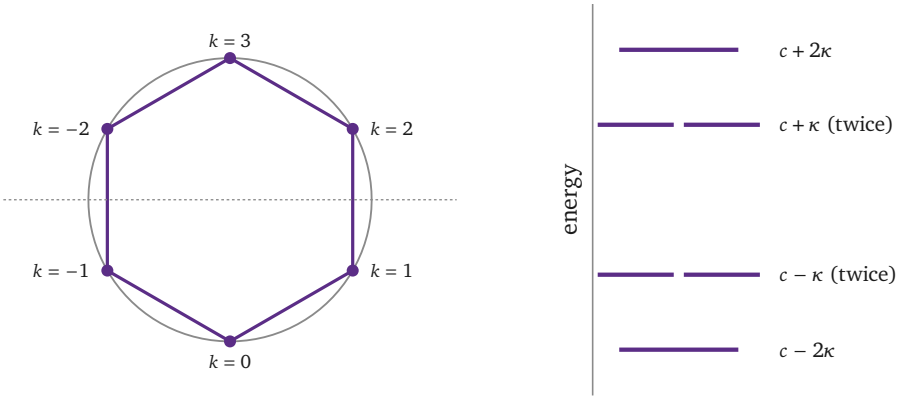


Figure 8.1: The levels of the ring of six sites (Example 8.2). Left: the energies $c - 2\kappa\cos(2\pi k/6)$ are the heights of the vertices of a regular hexagon inscribed in a circle of radius 2κ, whose centre lies on the dotted line at the energy c, with one vertex at the bottom. Right: the four levels and their degeneracies.

and

$$\frac{1}{\sqrt{2}}(f_1 + f_3) = \frac{1}{\sqrt{2}}(e_0 - e_2),$$

a stationary state of energy c in which the particle is never found on the sites 1 and 3.

8.4 Benzene

Example 8.2 (Benzene in the Hückel model)**.** The benzene molecule is a ring of six carbon atoms. The model of Hückel treats each electron on its own, as a particle on the ring with one state for each carbon atom, an on-site energy c and a hopping amplitude κ between neighbours. Hops to farther atoms are neglected. This is (8.1) with $N = 6$. The energy eigenstates of one electron are called its *orbitals*. By Theorem 8.1, the energies of the six orbitals are

$$c - 2\kappa \ (k = 0), \qquad c - \kappa \ (k = \pm 1), \qquad c + \kappa \ (k = \pm 2), \qquad c + 2\kappa \ (k = 3),$$

two simple levels and two degenerate pairs (Figure 8.1). The model gives the orbitals and their order, not the measured spectrum of the molecule.

The hexagon in Figure 8.1 is the construction of Frost and Musulin. In it, the vertex at the angle $2\pi k/6$ from the bottom has the height $c - 2\kappa\cos(2\pi k/6)$. For any N, the same holds with a regular N-gon.

8.5 A particle started on one site

Prepare the particle on the site 0, in the state e_0, and let it hop. In the Fourier basis, $e_0 = N^{-1/2}\sum_k f_k$, since $\langle f_k, e_0\rangle = N^{-1/2}$. By Theorem 4.4, each f_k gains the phase $e^{-\mathrm{i}t\varepsilon_k/\hbar}$, so the amplitude to find the particle on the site x at the time t is

$$\langle e_x, U_t e_0\rangle = \frac{1}{N}\sum_{k\in\mathbb{Z}_N} \zeta^{kx} e^{-\mathrm{i}t\varepsilon_k/\hbar}. \tag{8.4}$$

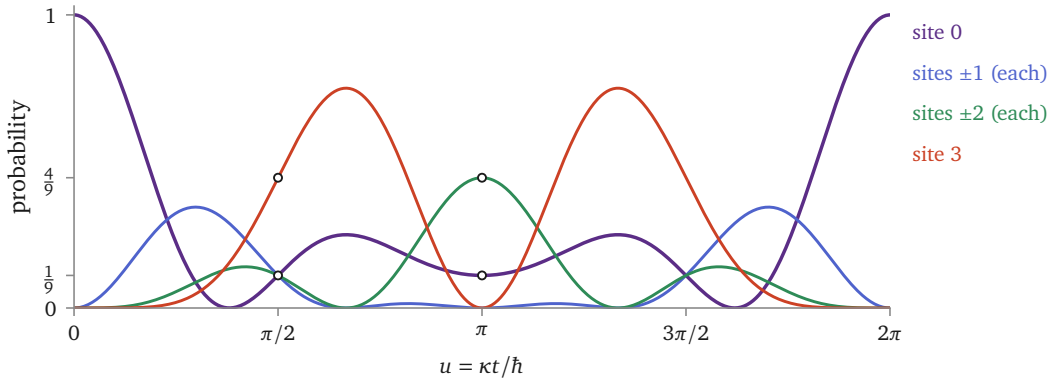


Figure 8.2: A particle started on the site 0 of the ring of six sites: the probability of each site against $u = \kappa t/\hbar$, from (8.4). The sites ±1 have equal probabilities, and so have the sites ±2, by the mirror. The circles mark $\frac{1}{9}$ and $\frac{4}{9}$ at $u = \pi/2$ and $u = \pi$.

For $N = 6$, put $u = \kappa t/\hbar$. The six energies are $c - 2\kappa$, $c - \kappa$ twice, $c + \kappa$ twice and $c + 2\kappa$, so on the site 0

$$\langle e_0, U_t e_0 \rangle = \frac{e^{-\mathrm{i}ct/\hbar}}{6}\left(e^{2\mathrm{i}u} + 2e^{\mathrm{i}u} + 2e^{-\mathrm{i}u} + e^{-2\mathrm{i}u}\right) = e^{-\mathrm{i}ct/\hbar}\,\frac{\cos 2u + 2\cos u}{3}.$$

Figure 8.2 shows all six probabilities. At $u = \pi/2$ the particle is on the opposite site, $x = 3$, with probability $\frac{4}{9}$, and on each other site with probability $\frac{1}{9}$. At $u = \pi$ it is on each of the sites $x = \pm 2$ with probability $\frac{4}{9}$ and back on the site 0 with probability $\frac{1}{9}$. At $u = 2\pi$, that is, $t = 2\pi\hbar/\kappa$, every phase $e^{-\mathrm{i}t\varepsilon_k/\hbar}$ equals $e^{-\mathrm{i}ct/\hbar}$, since the differences $\varepsilon_k - c$ are whole multiples of κ. So $U_t e_0 = e^{-\mathrm{i}ct/\hbar} e_0$, and the particle is back on the site 0, with certainty. The return is exact because the Bohr frequencies of the ring of six sites are whole multiples of $\kappa/\hbar$. For most N they are not, and the particle never returns exactly (Exercise Sheet 8).

The mirror explains the equal pairs in Figure 8.2. It is a symmetry and leaves e_0 fixed, so $MU_t e_0 = U_t M e_0 = U_t e_0$. Since M is unitary and $Me_x = e_{-x}$, the amplitudes on the sites x and $-x$ agree.

8.6 A ring with a flux

For a charged particle, a magnetic field through the middle of the ring is described, in a simple model, by a phase on each hop. With the phase $e^{\mathrm{i}\theta}$ on each hop forward, θ real, the Hamiltonian is

$$H_\theta = c\mathbb{1} - \kappa\left(e^{\mathrm{i}\theta}T + e^{-\mathrm{i}\theta}T^\dagger\right).$$

The phase of a whole loop round the ring is $N\theta$. The shift still commutes with H_θ, and the same calculation as in the proof of Theorem 8.1 gives $H_\theta f_k = \varepsilon_k(\theta) f_k$ with

$$\varepsilon_k(\theta) = c - 2\kappa\cos\left(\theta - \frac{2\pi k}{N}\right).$$

Two of these are equal, $\varepsilon_k(\theta) = \varepsilon_l(\theta)$ with $k \neq l$, only when $\theta - 2\pi k/N = -(\theta - 2\pi l/N)$ up to a multiple of 2π, that is, when $\theta = \pi(k + l)/N$ up to a multiple of π. So for θ not a multiple of π/N every level is simple.

The mirror is no longer a symmetry, since $MH_\theta M = H_{-\theta}$, which differs from H_θ unless θ is a multiple of π. The shift remains, and alone it forces nothing, as Lecture 6 noted after

Theorem 6.5. At the special values $\theta = \pi r/N$, for an integer r, a level is degenerate again, and another symmetry, a mirror combined with a phase on each site, appears (Exercise Sheet 8).

The physics

Hückel proposed the model of Example 8.2 in 1931 to explain the stability of benzene. The model keeps six electrons, one from each carbon atom, which chemists call the π electrons. In the lowest energy state of the molecule, in this model, they fill the three lowest orbitals, $k = 0$ and $k = \pm 1$, two electrons to an orbital. Why at most two is the subject of Lecture 14. Frost and Musulin gave the polygon construction of Figure 8.1 in 1953. A magnetic flux through a ring of metal about a micrometre across shifts the energies of its electrons as in § 8.6. In a cold ring a small current can then circulate in equilibrium, with no battery. Such persistent currents have been measured in rings of aluminium (Bleszynski-Jayich et al., 2009). Hückel wrote the hopping as β, with $\beta < 0$. Here $\beta = -\kappa$. In a long chain of carbon atoms with alternating single and double bonds, polyacetylene, the model of § 8.7 gives a gap between two bands. Su, Schrieffer and Heeger used such a model in 1979. Doped with other atoms, the polymer conducts, and the Nobel prize in chemistry of 2000 went to Heeger, MacDiarmid and Shirakawa for conducting polymers.

8.7 Two sites in each cell

In a chain of carbon atoms joined alternately by single and double bonds, the hops alternate between two values. The shift T by one site is then no longer a symmetry, but the shift T^2 by two sites still is. Take a ring of $2N$ sites, $N \geq 2$, with $\zeta = e^{2\pi \mathrm{i}/2N}$, the ζ of this ring, and read it as N *cells*. The cell $x \in \mathbb{Z}_N$ holds the sites $2x$ and $2x+1$. Let the hop inside a cell be $\kappa_1 > 0$ and the hop between cells $\kappa_2 > 0$:

$$H = c\mathbb{1} - \sum_{x \in \mathbb{Z}_N} \Big(\kappa_1 \big(|e_{2x+1}\rangle\langle e_{2x}| + |e_{2x}\rangle\langle e_{2x+1}| \big) + \kappa_2 \big(|e_{2x+2}\rangle\langle e_{2x+1}| + |e_{2x+1}\rangle\langle e_{2x+2}| \big) \Big). \tag{8.5}$$

For $k \in \mathbb{Z}_N$ and $v = (v_a, v_b) \in \mathbb{C}^2$, the *Bloch wave*

$$g_k(v) = \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} \zeta^{2kx} \big(v_a e_{2x} + v_b e_{2x+1} \big)$$

repeats the pattern v in every cell, with the phase advancing by ζ^{2k} from cell to cell.

Theorem 8.3 (Bloch's theorem on a ring)**.** *For each $k \in \mathbb{Z}_N$, the vectors $g_k(v)$, $v \in \mathbb{C}^2$, form the eigenspace $\mathcal{B}_k$ of T^2 for the eigenvalue ζ^{-2k}, and this eigenspace is a plane. The space $\mathbb{C}^{2N}$ is the orthogonal sum of the N planes $\mathcal{B}_k$. Every H that commutes with T^2 maps each $\mathcal{B}_k$ into itself, so $Hg_k(v) = g_k(h(k)v)$ for a 2×2 matrix $h(k)$, and the levels of H are the eigenvalues of the N matrices $h(k)$.*

Proof. As in § 8.2, $T^2 g_k(v) = \zeta^{-2k} g_k(v)$. By the geometric sum over x, the map $v \mapsto g_k(v)$ is linear and keeps norms, so $\mathcal{B}_k$ is a plane. The same sum makes $\mathcal{B}_k$ and $\mathcal{B}_l$ orthogonal for $k \neq l$ in $\mathbb{Z}_N$. The N planes fill $\mathbb{C}^{2N}$, since their dimensions add up to $2N$. The numbers ζ^{-2k} are distinct, so each $\mathcal{B}_k$ is a whole eigenspace of T^2. Since T^2 commutes with H, the argument of § 8.2 shows that H maps each eigenspace of T^2 into itself. ∎

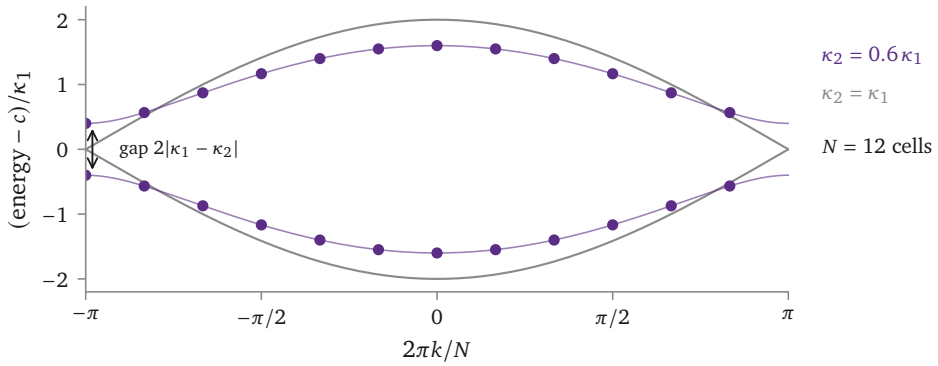


Figure 8.3: Two sites in each cell (Example 8.4): the levels $c \pm \left|\kappa_1 + \kappa_2 \zeta^{2k}\right|$ against $2\pi k/N$ for $N = 12$ cells and $\kappa_2 = 0.6\,\kappa_1$ (dots), with the curves they lie on. In grey, equal hops: the ring of $2N$ sites, whose two sets of levels meet at $k = N/2$.

So the $2N$ levels come from N matrices of size 2. The lower eigenvalues of the matrices $h(k)$ form one *band* of N levels, and the upper eigenvalues another. With p sites in each cell, the same proof gives N matrices of size p.

Example 8.4 (Alternating hops). Apply (8.5) to $g_k(v)$ and collect the coefficients of e_{2x} and e_{2x+1}:

$$h(k) = \begin{pmatrix} c & -(\kappa_1 + \kappa_2 \zeta^{-2k}) \\ -(\kappa_1 + \kappa_2 \zeta^{2k}) & c \end{pmatrix},$$

whose eigenvalues are, by Proposition 6.7,

$$c \pm \left|\kappa_1 + \kappa_2 \zeta^{2k}\right|, \qquad \left|\kappa_1 + \kappa_2 \zeta^{2k}\right|^2 = \kappa_1^2 + \kappa_2^2 + 2\kappa_1\kappa_2 \cos\frac{2\pi k}{N}.$$

These are the two bands (Figure 8.3). For $\kappa_1 = \kappa_2 = \kappa$ they are $c \pm 2\kappa\,|\cos(\pi k/N)|$, the levels of the ring of $2N$ sites in Theorem 8.1, sorted into two sets. For $\kappa_1 \neq \kappa_2$ every level of the upper band lies above every level of the lower band, by at least $2\,|\kappa_1 - \kappa_2|$, since $\left|\kappa_1 + \kappa_2 \zeta^{2k}\right| \geq |\kappa_1 - \kappa_2|$ by the triangle inequality. For even N the label $k = N/2$ reaches this bound. The interval between the two bands, which holds no level, is a *gap*. For benzene with alternating bonds, $N = 3$, and the levels are $c \pm (\kappa_1 + \kappa_2)$ and, twice each, $c \pm \sqrt{\kappa_1^2 - \kappa_1\kappa_2 + \kappa_2^2}$. The pairs survive because T^2 and the mirror through the middle of a bond, $e_y \mapsto e_{1-y}$, are symmetries that do not commute (Theorem 6.5).

A band on a finite ring is N levels, not a continuum. The continuum is the limit $N \to \infty$, taken in *Continuous Systems*. Equal hops give the ring of $2N$ sites, whose two sets of levels meet at $k = N/2$ for even N. The gap needs $\kappa_1 \neq \kappa_2$ to stay open as N grows.

A particle on a ring

States: the rays of $\mathbb{C}^N$, with the site basis e_x, $x \in \mathbb{Z}_N$, and the Fourier basis f_k, $k \in \mathbb{Z}_N$. *Observables:* those diagonal in the site basis, measured by the site reading, and the Hamiltonian $H = c\mathbb{1} - \kappa(T + T^\dagger)$. *Evolution:* f_k gains the phase $e^{-\mathrm{i}t\varepsilon_k/\hbar}$, with $\varepsilon_k = c - 2\kappa\cos(2\pi k/N)$. The levels of k and $-k \neq k$ coincide (Theorem 8.1). With two sites in each cell, the levels form two bands (Theorem 8.3). *Measurement:* the site reading of ψ gives x with probability $|\langle e_x, \psi\rangle|^2$.

Summary.

- *The levels of the ring*: the Fourier modes f_k diagonalise the shift and the Hamiltonian, with the energies $c - 2\kappa\cos(2\pi k/N)$. The degenerate pairs are k and $-k$ (Theorem 8.1).
- *Bloch's theorem on a ring*: a Hamiltonian that commutes with the shift by one cell of two sites has its levels in N matrices of size 2, two bands (Theorem 8.3).

For six sites the particle returns exactly, at $t = 2\pi\hbar/\kappa$. Lecture 9 reads the particle in the Fourier basis, and finds a pair of unitaries, one for each reading, that do not commute.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §4.3, pp. 265–269. Banks, *Quantum Mechanics: An Introduction*, §3.2.1, pp. 64–68. Hückel, *Zeitschrift für Physik* 70, 204 (1931). Frost and Musulin, *Journal of Chemical Physics* 21, 572 (1953). Bleszynski-Jayich et al., *Science* 326, 272 (2009). Su, Schrieffer and Heeger, *Physical Review Letters* 42, 1698 (1979).